PROBLEM 4 / UTILITIES

GridPilot

Choose energy actions as conditions change.

An autonomous renewable energy simulation coordinating solar, wind, batteries, industrial consumers and grid exchange - then validating and adapting.

15 minutes
between decisions

2 hours
planning horizon

96 decisions
per simulated day

Synthetic data. Simulated tools. Bounded model-based planning. No real grid connection.

THE OPERATING PROBLEM

Every interval changes the best action.

Renewable variability and operational limits turn a single dispatch decision into a sequence of trade-offs.

WEATHER + INPUTS

Clouds, wind surges and updated forecasts alter generation and reserve needs.

MARKET

Energy and carbon prices, demand incentives and curtailment charges shape trade-offs.

ASSETS + LOAD

Critical loads, battery outages, line caps and maintenance restrict available actions.

PORTFOLIO IN THE PROTOTYPE

5 solar farms + 3 wind farms | 2 batteries | 3 industrial consumers | 2 grid lines

Generation is aggregated for planning. Assets expose individual output and availability; loads retain separate ledgers.

AGENTIC DECISION INTELLIGENCE

Observe. Plan. Act. Check. Adapt.

An executable model searches feasible trajectories, takes a simulated action, and observes the result.

01 OBSERVE

Weather, market, consumers, asset health, line limits and maintenance.

02 ANALYZE

Requested and effective objectives; reserves and availability risks.

03 PLAN

Stage maintenance decisions in the forecast. Search eight steps with a 12-state beam.

04 EXECUTE

Prepare renewables, batteries, consumer, route, market and maintenance tools.

05 VALIDATE

Independent accounting, line caps, critical loads and job windows before commit.

06 ADAPT

Refresh forecasts and state; roll back and replan after a tool failure.

Receding horizon: search 2 hours ahead, execute the first 15-minute action, then observe again.

Make competing priorities visible.

Storms or large frequency-proxy excursions temporarily prioritize reliability. Inspect weights and alternatives.

WHAT THE PLANNER BALANCES

- \+ Trades, carbon charges and demand incentives
- \+ Unserved demand and uncertainty reserves
- \+ Solar / wind curtailment charges
- \+ Battery wear and a frequency-response proxy

WHAT EVERY DISPATCH MUST RESPECT

- \+ Battery SOC, power, efficiency and availability
- \+ Per-line capacity and source curtailment
- \+ Energy balance and critical-load priority
- \+ Consumer backlogs and maintenance deadlines

Units: MW, MWh, INR/MWh; carbon prices in INR/tonne. Grid carbon: 420 kg CO2/MWh. Frequency is an illustrative planning signal, not an AC grid-stability model. Search is bounded, without a global optimality guarantee.

EXECUTION + RECOVERY

A failed tool cannot leave a partial dispatch.

Simulated actions use staged, idempotent transactions with an auditable recovery path.

PREPARE

Stage dispatch, consumer and maintenance actions.

>

VALIDATE

Check energy, asset health, line caps and obligations.

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COMMIT

Apply one complete transaction atomically.

IF A TOOL FAILS

Rollback > Reobserve > Replan > Revalidate > Commit or record failure

The audit separates failed attempts from committed actions. Retry idempotency prevents the same transaction from applying twice.

All execution happens inside the simulation. No electricity trade or physical equipment is controlled.

Show decisions under changing conditions.

27 tests passed: 2,592 seeded scenario intervals and 384 mixed-shock intervals.
Separate evaluator: 993/993 cycles validated. Passing constraints does not mean every load was served.

INPUTS CHANGE DECISIONS

Adjust carbon price, demand incentives, cloud cover, wind or frequency proxy; inspect the resulting plan.

CRITICAL LOADS + MAINTENANCE

Inspect three consumer allocations and two line flows; show a maintenance schedule, delay or inspection receipt.

SHOCKS + FAILURE RECOVERY

Inject a wind surge, cloud, price spike or outage; force a tool rejection and inspect rollback and replanning.

DAY RUN + EXPORTED EVIDENCE

Simulate 96 intervals. Compare service and costs with the no-storage baseline; export the complete audit.

Exact checks and scenario results: VALIDATION.md. Coverage and assumptions: REQUIREMENTS_COVERAGE.md. Costs include initial inventory and different final SOC; report service and pending load. No annual savings or payback claim.

SCOPE + SUBMISSION CLAIM

Proposed grid position: F3 / D1

The final claim must match the working demo, architecture and evidence; the brief penalizes over- and underestimation.

FEATURE COVERAGE

F1: Coordinated, feasible action selection.

F2: Situation-dependent criteria and uncertainty reserve.

F3: Time simulation with changing input conditions.

DEPTH + LIMITS

D1: Structured synthetic inputs and demonstrated decisions.

D2 needs defensible reliability evidence. D3 is not claimed.

No live LLM, trained forecast, real grid connection or AC power flow. Health and frequency are simplified models.

Submission: working prototype and accessible repository, pitch PDF, 2-4 minute demo, detailed architecture and justified grid declaration. Confirm repository publication and portal submission separately. Brief: pp. 2-4, 10-14; portal screenshot: formats.